CMT-Transformer: A Probabilistic Task-Coupled Framework with Empirical Prior Integration for Cognitive Rehabilitation Decision Support

Abstract Cognitive rehabilitation decision support involves transforming heterogeneous assessment information into cognitive-status estimates and downstream rehabilitation-domain predictions. Existing multi-task approaches often treat related tasks as parallel objectives or rely on hard-label transfer, limiting their ability to represent asymmetric dependencies and probabilistic information flow. This study proposes CMT-Transformer, a probabilistic task-coupled framework for joint cognitive-status stratification and rehabilitation-domain prediction. The framework integrates self-supervised representation learning, Transformer-based feature modeling, probabilistic task conditioning, and empirical prior integration. Unlike hard cascades, CMT-Transformer transfers the complete cognitive-status probability distribution as task-level information, preserving relative support for alternative assessment states during downstream prediction. A training-derived assessment-rehabilitation prior is incorporated into the domain logits through a learnable fusion coefficient while retaining sample-specific representations. Experiments on a cohort of 5,099 older adults derived from the China Health and Retirement Longitudinal Study showed that CMT-Transformer achieved a mean accuracy of 0.8899 for cognitive-status stratification and a mean average precision of 0.9535 for rehabilitation-domain prediction, yielding the strongest mean performance across the reported metrics. Ablation studies, sensitivity analyses, statistical comparisons, and computational evaluations further characterized the contributions of probabilistic task coupling and empirical prior integration. An interactive web-based system was developed to demonstrate the complete inference workflow. The framework provides a practical approach for modeling directional dependence between cognitive assessment and rehabilitation-domain prediction in structured decision-support settings.

Keywords: Probabilistic task coupling; Multi-task learning; Tabular Transformer; Empirical prior integration; Cognitive-status stratification; Rehabilitation-domain prediction

1. Introduction

Population ageing has increased the need for scalable cognitive assessment and rehabilitation support. Machine-learning models have been used to identify community-dwelling older adults who may be at increased risk of cognitive impairment from routinely collected demographic, health, and lifestyle information [1]. Rehabilitation research has also expanded beyond conventional exercise delivery to technology-assisted assessment and intervention, including brain-computer interfaces, virtual reality, neurostimulation, and robot-assisted training [2]. Machine-learning-enabled robotic systems have further been investigated for functional capacity evaluation and occupational rehabilitation, where objective measurements may complement clinician-led assessment [3]. These developments reflect a broader shift from isolated prediction tools toward computational systems that connect assessment information with subsequent rehabilitation planning.

Artificial intelligence has been applied at different stages of this process. An individualized treatment-response model has been developed to estimate the likely effects of pharmacological interventions for cognitive impairment in dementia [4]. In stroke rehabilitation, multimodal machine learning has been used to predict Fugl-Meyer scores and motor recovery potential from heterogeneous clinical information [5]. Transformer and convolutional representations have also been combined for posture estimation in stroke-oriented health monitoring [6]. Although these studies involve different populations and clinical objectives, they illustrate the potential value of transforming heterogeneous observations into information that can assist assessment, prognosis, and rehabilitation planning.

Cognitive rehabilitation decision support introduces a related but distinct modeling problem. Cognitive-status stratification and rehabilitation-domain prediction are associated, but they do not have equivalent roles. An estimated cognitive state can provide relevant context for determining which rehabilitation domains may be applicable. The reverse relationship is less direct because a rehabilitation-domain label does not determine the original cognitive-status category. The two tasks therefore exhibit a directional rather than purely symmetric relationship.

This distinction is not fully represented by a conventional multi-task model with a shared encoder and parallel prediction heads. Parameter sharing may help the tasks learn from common input characteristics, but it does not explicitly pass the assessment result to the rehabilitation-domain branch. A hard cascade makes the direction of information flow visible, yet reduces the upstream output to a single predicted category. Any early classification error may then be transferred directly to the downstream task, while the relative support for alternative cognitive states is discarded. A more suitable formulation should retain the complete upstream distribution and allow it to participate in joint optimization.

Reliable representation learning is another concern. Expert-endorsed rehabilitation-domain labels require specialist input and are more difficult to obtain than unannotated assessment profiles. Self-supervised learning offers a practical way to use such profiles before task-specific fine-tuning and has been studied

across electronic health records, medical images, physiological signals, and other biomedical data [7]. SimCLR demonstrated that representations can be learned by maximizing agreement between augmented views, with augmentation design and the projection head having a substantial effect on downstream performance [8]. Studies in medical-image classification further show that the value of self-supervised pre-training depends on the characteristics of the data, the pre-training objective, and its relationship to the downstream task [9]. Momentum Contrast introduced a momentum-updated encoder and a queue-based representation dictionary, allowing a large and comparatively consistent set of negative samples to be maintained during training [10]. These studies establish useful principles for encoder initialization, although they do not address the relationship between downstream tasks.

The input representation must also accommodate heterogeneous structured variables. Attention-based architectures have increasingly been adapted to table-related problems. TableFormer introduces structure-aware attention biases for table-text encoding and reduces sensitivity to arbitrary row and column arrangements [11]. SAINT applies attention across feature and sample dimensions and combines the resulting tabular architecture with contrastive pre-training [12]. FT-Transformer represents numerical and categorical variables as feature-specific tokens and applies self-attention to their interactions [13]. Its comparative evaluation also shows that neural tabular models can be competitive, although no architecture is consistently superior across all datasets. These findings support feature tokenization and self-attention for structured cognitive profiles, but they do not determine how an estimated assessment state should be used by a related downstream task.

Multi-task learning (MTL) provides a natural starting point for joint prediction. In a rehabilitation-oriented sensing application, motion intention and muscle-fatigue level have been learned simultaneously from surface electromyography, demonstrating that related outputs can benefit from a common representation [14]. The present problem introduces an additional requirement. Cognitive status is not simply another output placed alongside rehabilitation-domain prediction; it provides an intermediate assessment state that is relevant to the downstream branch. The model should therefore combine shared representation learning with an explicit, differentiable pathway from assessment to rehabilitation-domain prediction.

Based on this formulation, this study proposes CMT-Transformer, a probabilistic task-coupled framework for joint cognitive-status stratification and rehabilitation-domain prediction. A feature-tokenized Transformer encoder is initialized through masked feature reconstruction and contrastive pre-training. During supervised fine-tuning, the complete cognitive-status probability vector is combined with the shared representation to condition rehabilitation-domain prediction. No hard category is inserted into this pathway, allowing the downstream branch to retain the relative probabilities assigned to all cognitive-status classes.

A cross-task prior is additionally estimated from the co-occurrence of cognitive-status categories and expert-endorsed rehabilitation domains in the training partition. The prior is projected through the predicted

cognitive-status distribution and incorporated into the downstream logits through a learnable coefficient. It therefore provides population-level guidance without replacing the sample-specific output of the conditional prediction branch. The resulting rehabilitation-domain scores are intended to support domain-level assessment and should not be interpreted as direct treatment prescriptions. The main contributions are summarized as follows:

- A probabilistic task-coupled learning framework is developed for structured cognitive decision systems. The proposed framework formulates cognitive status stratification and rehabilitation-domain prediction as asymmetric but related tasks, enabling information transfer through probabilistic task-level representations rather than conventional hard-label cascading.
- An assessment-conditioned multi-task architecture built on a feature-tokenized Transformer is proposed to integrate representation learning and explicit task-dependency modeling. By combining self-supervised representation learning with probabilistic conditioning, the framework captures latent relationships between assessment states and downstream decision objectives.
- A learnably weighted empirical prior integration mechanism is proposed to incorporate training-derived cross-task associations into the learning process. The proposed strategy provides additional guidance from population-level relationships while retaining the flexibility of individual representation learning.
- Comprehensive experiments are conducted on a CHARLS-derived cohort to evaluate the proposed framework. Experimental comparisons, ablation studies, sensitivity analyses, statistical evaluations, and computational assessments demonstrate the effectiveness of the proposed approach for intelligent cognitive decision support. An interactive web-based system was developed to demonstrate the complete inference workflow.

The remainder of this paper is organized as follows. Section 2 reviews related studies. Section 3 introduces the problem formulation. Section 4 presents the proposed CMT-Transformer framework. Section 5 reports the experimental results and analysis. Finally, Section 6 concludes this study.

2. Related Work

2.1 Self-supervised representation learning

Expert-endorsed labels are usually more costly to obtain than unannotated assessment profiles. Self-supervised learning provides a way to use the latter before supervised fine-tuning. In medicine and healthcare, self-supervised methods have been studied for electronic health records, medical images, physiological signals, and other biomedical data [7]. Their training objectives generally rely on recovering altered inputs or learning consistent representations from different views of the same observation.

SimCLR learns representations by maximizing agreement between augmented views and shows that augmentation design, the projection head, batch size, and training duration can substantially affect representation quality [8]. A systematic review of self-supervised medical-image classification similarly

reports that performance depends on the pre-training task, the source data, and the correspondence between pre-training and downstream evaluation [9]. Momentum Contrast follows a different implementation strategy, using a momentum-updated encoder and a queue to maintain a relatively large and stable dictionary of negative representations [10].

These methods were largely developed in visual or broader biomedical settings. Their use with structured cognitive profiles requires feature-level perturbations that do not assume image geometry. Masked reconstruction can encourage the encoder to infer missing assessment variables from the remaining context, whereas contrastive learning can improve consistency across perturbed versions of the same profile. The two objectives may therefore provide complementary initialization. They improve how the input is represented, but do not define how one downstream output should influence another.

More specifically, for structured tabular data, VIME combines mask estimation with feature reconstruction [15], while SCARF constructs contrastive views through random feature corruption [16]. Compared with image-oriented augmentation strategies, these approaches are more compatible with heterogeneous assessment profiles containing mixed demographic and cognitive variables. However, similar to other representation-learning methods, they primarily focus on improving feature quality rather than modeling directional dependencies between multiple downstream tasks.

2.2 Transformer models for structured data

Transformer architectures have gradually been extended from conventional sequence modeling to table understanding and tabular prediction. TableFormer introduces learnable structural biases into the attention calculation for joint table-text encoding [11]. Its design reduces sensitivity to arbitrary row and column order, although the task differs from clinical tabular classification.

SAINT applies attention across both columns and rows and combines the architecture with contrastive pre-training for tabular data [12]. FT-Transformer represents each numerical or categorical variable with a feature-specific token and uses self-attention to model interactions among these tokens [13]. The comparative study accompanying FT-Transformer also gives a more qualified view of tabular deep learning: carefully designed neural architectures can be strong competitors, but tree-based models and neural models each retain advantages on different datasets.

TransTab further investigates transferable Transformer representations across heterogeneous tables with different schemas and demonstrates the potential of pre-trained tabular models for cross-table prediction [17]. Although these Transformer-based approaches improve feature interaction modeling, they generally generate task outputs from shared representations without explicitly incorporating the probabilistic relationship between an upstream assessment task and a downstream decision task.

Structured cognitive profiles contain mixed variable types, no natural sequential order, and relationships that may extend across several assessment items. Feature tokenization preserves the identity of each variable, while self-attention allows interactions to be learned without imposing a fixed neighborhood

structure. These properties make Transformer encoders suitable for the present input format.

The output structure remains a separate issue. Most tabular Transformers are designed to produce one target or several outputs from a common representation. When several heads are attached in parallel, their relationship is represented mainly through the shared encoder. An upstream assessment result is not explicitly made available to the downstream branch.

2.3 Multi-task learning and output dependency

Multi-task learning uses common parameters or representations to exploit information across related objectives. In rehabilitation-oriented sensing, motion-intention recognition and muscle-fatigue classification have been jointly modeled from surface electromyography [14].

MMoE uses shared experts and task-specific gating networks to learn different representations for related tasks [18]. However, its task outputs remain parallel, and the downstream task does not explicitly receive the complete probability distribution produced by another task. FineRehab provides a multimodal, multi-task dataset for rehabilitation analysis and illustrates the diversity of observations and targets that may occur within the same rehabilitation setting [19].

Parallel multi-task architectures are useful when tasks share information but have no clear ordering. Other designs introduce a more explicit dependency. The joint many-task model of Hashimoto et al. arranges several natural-language processing tasks hierarchically and transfers intermediate representations from lower-level to higher-level tasks [20]. Classifier chains provide previously predicted labels as additional inputs to later classifiers, thereby modeling dependencies among multi-label outputs [21]. Prototype-aware structured contrastive distillation has also been used to retain semantic relationships among multiple labels [22].

These approaches show that later predictions can benefit from earlier task outputs rather than relying only on a common encoder. Hard sequential prediction, however, introduces a practical limitation. Once an upstream distribution is converted into one category, the alternative states and their relative probabilities are lost. An incorrect upstream decision may also be propagated directly to the next stage.

Cognitive-status stratification and rehabilitation-domain prediction have this type of directional relationship. Cognitive status provides context for the downstream domain prediction, but rehabilitation-domain labels do not determine the assessment category. CMT-Transformer therefore transfers the full cognitive-status probability vector. The vector is treated as a differentiable input to the downstream branch, not as a final hard decision.

2.4 Prior-informed learning and task-specific objectives

Prior information has been introduced into deep models when learned features alone may not fully represent stable structural patterns. Prior-guided reconstruction has been studied for accelerated multi-contrast magnetic resonance imaging [23]. Prior-guided feature enhancement has been used in few-shot medical-image classification [24], and prior-guided feature enrichment has been developed for

few-shot segmentation [25].

These methods address imaging problems rather than cognitive assessment, but they illustrate several ways in which prior information can complement a learned representation. They also show why the prior should not simply replace the model output. A fixed or overly dominant prior can suppress sample-specific variation.

In the present setting, the available expert annotations contain recurring associations between cognitive-status categories and rehabilitation domains. These associations are estimated from the training partition and represented as a cross-task co-occurrence prior. The matrix is therefore data-derived; it is not an external clinical rule or a treatment guideline. Because individuals with the same cognitive-status category may still differ in their domain-level needs, the contribution of this prior is controlled by a learnable coefficient.

The two tasks also place different demands on optimization. Label smoothing and label relaxation can reduce excessive confidence in multiclass prediction [26]. Asymmetric loss is designed for multi-label settings in which numerous easy negatives may dominate the objective [27]. Bayesian personalized ranking optimizes the relative order of positive and negative candidates and is useful when ranked outputs matter in addition to independent label probabilities [28].

Within CMT-Transformer, these losses have supporting rather than independent methodological roles. Label-smoothed cross-entropy is used for cognitive-status discrimination, asymmetric loss addresses imbalance in rehabilitation-domain labels, and the ranking term encourages applicable domains to appear above non-applicable ones.

2.5 Limitations of existing approaches

The reviewed studies address several components of the current problem. Self-supervised learning provides encoder initialization when annotations are limited. Tabular Transformers model interactions among heterogeneous variables. Multi-task learning allows related objectives to share representations, and cascaded or chain-based methods introduce explicit output dependencies. Prior-guided models show how structural information may be added to data-driven prediction.

What remains insufficiently addressed is their combination under a directional, probabilistic task relationship. A parallel multi-task architecture does not explicitly use the estimated cognitive state in rehabilitation-domain prediction. A hard cascade does so, but discards the full class distribution. Prior-guided methods commonly incorporate information into one task, rather than jointly using an upstream probability distribution and a cross-task co-occurrence prior.

For the present problem, the downstream prediction should use cognitive-status information without assuming that the estimated category is certain. At the same time, population-level associations should guide, but not determine, the individual output. CMT-Transformer addresses these requirements by conditioning the downstream branch on the complete cognitive-status probability vector and separately

integrating a training-derived prior through a learnable weight. **Table 1** summarizes the main differences between representative research directions and the proposed framework.

Table 1 Comparison of representative research directions and the proposed framework.

Research direction	Primary focus	Treatment of task relationships	Intermediate task output	Use of prior information
Self-supervised methods	Encoder initialization and representation learning	Not usually considered	None	None
Transformer-based tabular methods	Interaction modeling among structured features	Usually handled through separate downstream heads	None	None
Parallel multi-task models	Shared representation across related tasks	Implicit through parameter sharing	Not directly transferred	Usually none
Cascaded or classifier-chain models	Explicit sequential dependency	Later predictions depend on earlier outputs	Commonly hard labels or decisions	Usually none
Prior-informed models	Incorporation of structural or domain-related guidance	Depends on the target task	Not central to the design	Fixed, estimated, or learned prior
CMT-Transformer	Joint representation learning and assessment-conditioned prediction	Explicit direction from cognitive-status assessment to domain prediction	Complete probability distribution	Training-derived cross-task prior with a learnable weight

The distinction is therefore not that CMT-Transformer uses more components than previous methods. Its main design choice is to preserve the upstream assessment distribution as part of downstream prediction and to combine that sample-level signal with a separately estimated population-level prior.

3. Problem Formulation

3.1 Problem definition

Cognitive-status stratification and rehabilitation-domain prediction are formulated as a joint learning problem. Let $\mathcal{D}=\left\{\left(\mathbf{x}_i, y_i^{\mathrm{cls}}, \mathbf{y}_i^{\mathrm{rec}}\right)\right\}_{i=1}^{N}$ denote a dataset containing N samples. For the i-th sample, $x_i \in \mathbb{R}^d$ is a structured assessment profile containing d demographic and item-level cognitive assessment features. The corresponding cognitive-status label is denoted by:

$$y_i^{\mathrm{cls}} \in \{1,2,\ldots,C\}$$

where C is the number of cognitive-status categories. The rehabilitation-domain annotations are represented by $\mathbf{y}_i^{\mathrm{rec}} \in \{0,1\}^R$, R is the number of candidate rehabilitation domains. The element $y_{ir}^{\mathrm{rec}}=1$ indicates that the r-th rehabilitation domain is considered applicable to the i-th sample.

These labels are expert-endorsed domain annotations derived from the available structured assessment profiles. They do not represent observed treatment allocation, intervention response, or clinically verified therapeutic benefit. Accordingly, the downstream task is defined as multi-label prediction of potentially applicable rehabilitation domains rather than direct treatment prescription.

The objective is to learn a cognitive-status classification function f_{cls} and a rehabilitation-domain prediction function f_{rec} :

$$\mathbf{p}_i^{\mathrm{cls}} = f_{\mathrm{cls}}(\mathbf{x}_i), \qquad \hat{\mathbf{y}}_i^{\mathrm{rec}} = f_{\mathrm{rec}}\left(\mathbf{x}_i, \mathbf{p}_i^{\mathrm{cls}}\right) \tag{1}$$

where $\mathbf{p}_i^{\mathrm{cls}} \in [0,1]^C$, $\sum_{c=1}^{C} p_{ic}^{\mathrm{cls}} = 1$ is the predicted probability distribution over the cognitive-status categories,

and $\hat{\mathbf{y}}_i^{\text{rec}} \in [0,1]^R$ is the predicted probability vector over the rehabilitation domains.

3.2 Probabilistic task-coupled formulation

AI- and machine-learning-based rehabilitation systems often involve multiple related analytical domains and prediction objectives [29]. Multi-task learning has also been used in medicine recommendation to combine heterogeneous attributes and task-related information within a joint learning framework [30]. In the present study, however, cognitive-status stratification and rehabilitation-domain prediction are not treated as fully symmetric tasks.

Cognitive status serves as an upstream assessment result that can inform the downstream domain prediction. The reverse relation is less direct, because a predicted rehabilitation domain does not determine the original cognitive-status category. The problem is therefore formulated as a probabilistic task-coupling process rather than as two parallel outputs connected only through shared parameters.

A shared encoder first maps the structured assessment profile to a latent representation:

$$\mathbf{h}_i = f_\theta(\mathbf{x}_i) \tag{2}$$

where f_θ denotes the encoder parameterized by θ.

The cognitive-status classification head then produces a categorical probability distribution:

$$\mathbf{p}_i^{\text{cls}} = \text{softmax}\left(g_{\text{cls}}(\mathbf{h}_i)\right) \tag{3}$$

where g_{cls} denotes the cognitive-status classification head.

The downstream rehabilitation-domain prediction is conditioned on both the shared representation and the cognitive-status probability vector:

$$\hat{\mathbf{y}}_i^{\text{rec}} = f_{\text{rec}}\left(\mathbf{h}_i, \mathbf{p}_i^{\text{cls}}\right) \tag{4}$$

Unlike a hard cascade, Eq. (4) does not reduce the upstream output to a single predicted class. The complete probability vector retains the relative evidence assigned to all cognitive-status categories and is used as a distribution-valued intermediate state. For samples close to a decision boundary, the downstream branch can therefore receive information about competing cognitive-status states rather than only the category with the largest probability.

The conditioning pathway also remains differentiable during supervised optimization. Consequently, the rehabilitation-domain objective can update the shared encoder and the assessment-conditioned prediction branch through backpropagation. This design differs from post-processing rules in which an independently generated cognitive-status label is inserted into a separate recommendation procedure after model training.

The probability vector should not be interpreted as a calibrated clinical uncertainty estimate. Its role in the present framework is to preserve the model's relative support for alternative cognitive-status categories and to provide a continuous task-level signal for downstream prediction.

3.3 Stage-wise learning objectives

CMT-Transformer is optimized in two sequential stages: self-supervised encoder initialization and supervised task-coupled fine-tuning. Self-supervised learning constructs training signals from the input data and can improve representation learning when downstream annotations are limited [31]. Contrastive objectives further encourage related views of the same sample to obtain similar representations while separating representations from different samples [32, 33].

During self-supervised pre-training, masked feature reconstruction and contrastive representation learning are jointly optimized:

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{recon}} + \lambda_c \mathcal{L}_{\text{contrast}} \tag{5}$$

where L_{recon} denotes the masked feature reconstruction loss, L_{contrast} denotes the contrastive loss, and $\lambda_c \geq 0$ controls the contribution of the contrastive objective.

The reconstruction term requires the encoder to infer masked variables from the remaining assessment context. The contrastive term encourages representation consistency between independently perturbed views of the same structured profile. These two objectives provide complementary supervision: one emphasizes the preservation of cross-variable information, whereas the other regularizes the latent space across feature-level perturbations.

During supervised fine-tuning, cognitive-status classification, multi-label rehabilitation-domain prediction, and domain ranking are optimized jointly:

$$\mathcal{L}_{\text{fine}} = \alpha \mathcal{L}_{\text{cls}} + \beta \mathcal{L}_{\text{ml}} + \delta \mathcal{L}_{\text{rank}} \tag{6}$$

where $\mathcal{L}_{\text{cls}}$ is the classification loss. $\mathcal{L}_{\text{ml}}$ is the multi-label prediction loss. $\mathcal{L}_{\text{rank}}$ is the ranking optimization term. The non-negative coefficients α, β, δ control the relative contributions of the three objectives.

The three terms address different aspects of the joint problem. The classification term separates cognitive-status categories, the multi-label term estimates the applicability of each rehabilitation domain, and the ranking term encourages applicable domains to receive higher scores than non-applicable domains. Their detailed definitions are provided in Section 4.

The encoder learned during the first stage is transferred to the downstream task-coupled framework before supervised optimization. Transferable deep models have previously been investigated for predicting rehabilitation outcomes across different training settings [34], while transfer learning more generally provides a means of reusing representations acquired from a source learning process in a related target task [35]. In CMT-Transformer, the source and target stages use the same structured profiles but different supervisory signals: the first stage learns from input-derived objectives, whereas the second stage adapts the representation to cognitive-status and rehabilitation-domain labels.

4. CMT-Transformer

4.1 Overview

CMT-Transformer is trained in two sequential phases: self-supervised encoder initialization and assessment-conditioned joint fine-tuning.

In the first phase, masked feature reconstruction and contrastive representation learning are optimized using input profiles from the training partition. The reconstruction objective requires the encoder to recover masked variables from the remaining feature context, whereas the contrastive objective encourages consistent representations for two perturbed views of the same sample.

In the second phase, the pre-trained encoder is transferred to the joint cognitive-status stratification and rehabilitation-domain prediction framework. The rehabilitation-domain branch receives both the shared Transformer representation and the predicted cognitive-status probability vector. A training-derived assessment–rehabilitation prior is then projected through the cognitive-status distribution and incorporated into the domain-prediction logits through a learnable coefficient.

Because the two phases are performed sequentially rather than optimized as a single end-to-end process, the phase-specific objective is written as:

$$\mathcal{L} = \begin{cases} \mathcal{L}_{\text{pre}}, & \text{during self-supervised pre-training} \\ \mathcal{L}_{\text{fine}}, & \text{during supervised fine-tuning} \end{cases} \tag{7}$$

The encoder obtained during pre-training initializes the shared Transformer backbone before supervised task adaptation. The complete architecture is shown in **Fig. 1**. It contains a feature-tokenized Transformer encoder, a cognitive-status classification branch, an assessment-conditioned rehabilitation-domain branch, and an empirical prior-integration module.

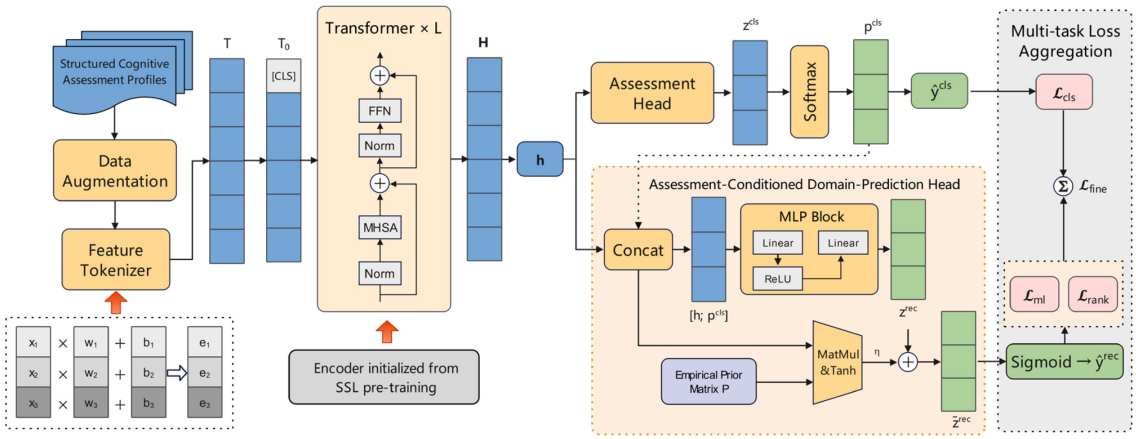


Fig. 1 Overall architecture of CMT-Transformer during assessment-conditioned multi-task fine-tuning.

4.2 Self-supervised encoder initialization

4.2.1 Masked feature reconstruction

For an input sample $\mathbf{x} \in \mathbb{R}^d$, a subset of feature dimensions is randomly selected and masked to generate a corrupted input $\tilde{\mathbf{x}}$. The encoder-decoder pair reconstructs the original profile as:

$$\hat{\mathbf{x}} = g_\psi\left(f_\theta\left(\tilde{\mathbf{x}}\right)\right) \tag{8}$$

where f_θ and g_ψ denote the encoder and reconstruction decoder, respectively. The parameters of the encoder and decoder are represented by θ and ψ, respectively.

Let $\mathcal{M}$ denote the set of masked feature indices. The reconstruction objective is evaluated only on the masked dimensions:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{M}|} \sum_{k \in \mathcal{M}} \left(x_k - \hat{x}_k \right)^2 \tag{9}$$

Restricting the loss to the masked dimensions prevents the model from being rewarded for simply copying the observed variables. Instead, the encoder must infer missing values from the available assessment context, which encourages it to retain dependencies among structured variables.

4.2.2 Contrastive representation learning

Two independently perturbed views of the same input profile are generated as:

$$\tilde{\mathbf{x}}^{(1)} = t_1(\mathbf{x}), \quad \tilde{\mathbf{x}}^{(2)} = t_2(\mathbf{x}), \quad t_1, t_2 \sim \mathcal{T} \tag{10}$$

where $\mathcal{T}$ denotes the predefined family of feature-level perturbations.

The two views are passed through the same encoder:

$$\mathbf{h}^{(1)} = f_\theta\left(\tilde{\mathbf{x}}^{(1)}\right), \quad \mathbf{h}^{(2)} = f_\theta\left(\tilde{\mathbf{x}}^{(2)}\right) \tag{11}$$

For a minibatch containing B samples, the contrastive objective is defined using the InfoNCE loss:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp\left(\text{sim}\left(\mathbf{h}_i^{(1)}, \mathbf{h}_i^{(2)}\right) / \tau\right)}{\sum_{j=1}^{B} \exp\left(\text{sim}\left(\mathbf{h}_i^{(1)}, \mathbf{h}_j^{(2)}\right) / \tau\right)} \tag{12}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity and $\tau > 0$ is the temperature coefficient. The representations generated from two augmented views of the same sample form a positive pair, whereas representations generated from different samples in the minibatch are treated as negative pairs.

Masked reconstruction and contrastive learning provide different forms of supervision. The former emphasizes recovery of feature relationships, whereas the latter regularizes the latent representation across input perturbations. They are jointly optimized through Eq. (5), and the resulting encoder parameters are transferred to the supervised stage.

4.3 Feature-tokenized Transformer encoder

The shared backbone uses a feature-tokenized Transformer to model interactions among heterogeneous structured variables. For the *j*-th input feature x_j, a feature-specific linear transformation is used to construct its token representation:

$$\mathbf{e}_j = x_j \mathbf{w}_j + \mathbf{b}_j, \quad \mathbf{e}_j \in \mathbb{R}^{d_{\text{model}}} \tag{13}$$

where $\mathbf{w}_j$ and $\mathbf{b}_j$ are trainable feature-specific parameters. A learnable classification token $\mathbf{e}_{\text{CLS}}$ is prepended to the feature-token sequence:

$$\mathbf{X}_0 = [\mathbf{e}_{\text{CLS}}; \mathbf{e}_1; \ldots; \mathbf{e}_d] \in \mathbb{R}^{(d+1) \times d_{\text{model}}} \tag{14}$$

Each variable has its own embedding parameters, allowing the model to preserve feature identity without depending on a conventional sequential positional encoding. This is appropriate for assessment profiles because the columns have semantic identities but do not form a natural temporal or linguistic sequence.

Within each Transformer layer, scaled dot-product attention is calculated as:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V} \tag{15}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices, respectively. d_k is the dimension of the attention key vector.

Several attention heads are computed in parallel and concatenated before linear projection. After the final Transformer layer, the token representations are denoted by:

$$\mathbf{H} \in \mathbb{R}^{(d+1)\times d_{\text{model}}} \tag{16}$$

The representation associated with the [CLS] token is extracted as the global sample representation:

$$\mathbf{h} = \mathbf{H}_{\text{CLS}} \in \mathbb{R}^{d_{\text{model}}}$$

This representation is shared by the cognitive-status branch and the assessment-conditioned rehabilitation-domain branch.

4.4 Assessment-conditioned rehabilitation-domain prediction

The cognitive-status classification branch first produces the multiclass logits and the corresponding probability distribution:

$$\mathbf{z}^{\text{cls}} = g_{\text{cls}}(\mathbf{h}), \quad \mathbf{p}^{\text{cls}} = \text{softmax}\left(\mathbf{z}^{\text{cls}}\right) \tag{17}$$

where $\mathbf{z}^{\text{cls}} \in \mathbb{R}^{C}$ contains the classification logits and $\mathbf{p}^{\text{cls}} \in [0,1]^{C}$ is the corresponding probability distribution.

Conventional multi-task models commonly place parallel task-specific heads after a shared encoder [36]. Such a design permits parameter sharing, but the downstream branch does not explicitly receive the output of the cognitive-status task.

CMT-Transformer introduces the predicted cognitive-status distribution as an additional input to the rehabilitation-domain branch. As shown in Fig. 2, the global representation and the complete probability vector are concatenated:

$$\mathbf{c} = [\mathbf{h};\mathbf{p}^{\text{cls}}] \tag{18}$$

The resulting conditional representation is mapped to the base rehabilitation-domain logits:

$$\mathbf{z}^{\text{rec}} = g_{\text{rec}}(\mathbf{c}), \quad \mathbf{z}^{\text{rec}} \in \mathbb{R}^{R} \tag{19}$$

No hard class decision is used in the conditioning pathway. The complete probability vector retains the relative probability assigned to each cognitive-status category. Because the probability vector remains differentiable during joint optimization, the rehabilitation-domain objective can update the shared encoder and the assessment-conditioned pathway.

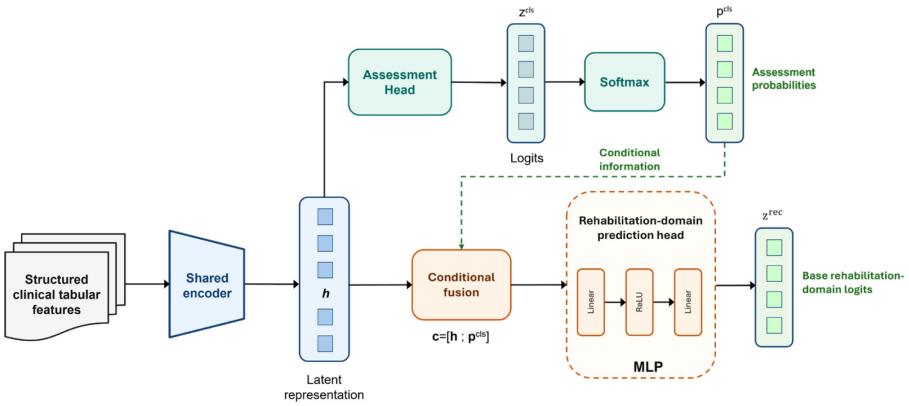


Fig. 2 Assessment-conditioned dependency from cognitive-status stratification to rehabilitation-domain prediction.

4.5 Empirical assessment-rehabilitation prior integration

Expert-endorsed annotations may contain recurring associations between cognitive-status categories and rehabilitation domains. To represent these population-level associations, an empirical prior matrix is constructed as:

$$\mathbf{P} \in [0,1]^{C \times R}$$

Each element P_{cr} represents the empirical probability that rehabilitation domain r is assigned to samples belonging to assessment category c. The matrix is calculated using only the training set:

$$P_{cr} = \frac{\sum_{i \in \mathcal{D}_{\text{train}}} \mathbb{I}\left(y_i^{\text{cls}} = c\right) y_{ir}^{\text{rec}}}{\sum_{i \in \mathcal{D}_{\text{train}}} \mathbb{I}\left(y_i^{\text{cls}} = c\right)} \tag{20}$$

where $\mathbb{I}(\cdot)$ is the indicator function. It represents empirical associations in the available annotations rather than an independently specified clinical rule, causal relation, or treatment guideline.

For each sample, the predicted cognitive-status distribution is projected into the rehabilitation-domain space through $\mathbf{P}$. As illustrated in **Fig. 3**, the projected prior signal is added to the base logits:

$$\tilde{\mathbf{z}}^{\text{rec}} = \mathbf{z}^{\text{rec}} + \eta \tanh\left(\mathbf{p}^{\text{cls}} \mathbf{P}\right) \tag{21}$$

where η is a learnable scalar fusion coefficient. The hyperbolic tangent function limits the magnitude of the prior signal and reduces the risk of excessive logit adjustment.

The final rehabilitation-domain probabilities are obtained as:

$$\hat{\mathbf{y}}^{\text{rec}} = \sigma\left(\tilde{\mathbf{z}}^{\text{rec}}\right) \tag{22}$$

where $\sigma(\cdot)$ is applied element-wise.

The matrix $\mathbf{P}$ remains fixed after being estimated from the training annotations. Adaptation is provided by the learnable coefficient η, while the assessment-conditioned prediction branch continues to produce sample-specific logits. The prior therefore acts as supplementary population-level guidance rather than a deterministic output rule.

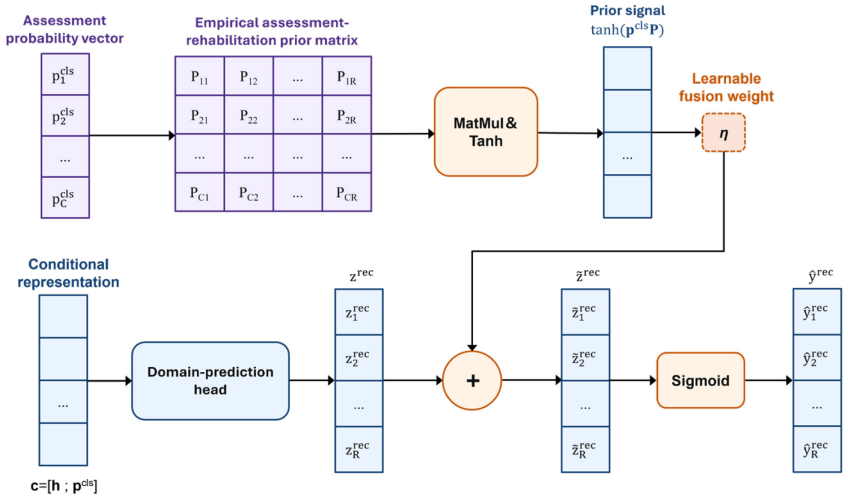


Fig. 3 Learnably weighted empirical assessment-rehabilitation prior-integration mechanism.

4.6 Joint optimization objectives

Label-smoothed cross-entropy is used for cognitive-status classification. Let $\varepsilon \in [0,1)$ denote the smoothing coefficient and let $\tilde{\mathbf{y}}_i^{\mathrm{cls}}$ be the smoothed target distribution. The classification loss is:

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C}\tilde{y}_{ic}^{\mathrm{cls}}\log p_{ic}^{\mathrm{cls}} \tag{23}$$

This objective reduces excessive confidence in cognitive-status predictions while retaining multiclass discrimination.

An asymmetric focal loss is used for multi-label rehabilitation-domain prediction:

$$\mathcal{L}_{\mathrm{ml}} = -\frac{1}{BR}\sum_{i=1}^{B}\sum_{r=1}^{R}\left[y_{ir}^{\mathrm{rec}}(1-\hat{y}_{ir}^{\mathrm{rec}})^{\gamma_+}\log\hat{y}_{ir}^{\mathrm{rec}} + (1-y_{ir}^{\mathrm{rec}})(\hat{y}_{ir}^{\mathrm{rec}})^{\gamma_-}\log(1-\hat{y}_{ir}^{\mathrm{rec}})\right] \tag{24}$$

where $\gamma_+ \geq 0$ and $\gamma_- \geq 0$ control the focusing strengths for positive and negative labels. Different focusing parameters are used because negative labels are usually more numerous and may otherwise dominate the objective.

A pairwise ranking loss is additionally used to improve the relative ordering of applicable and non-applicable domains. Let $\mathcal{P}$ denote the set of valid triplets (i, r^+, r^-), where r^+ is a positive rehabilitation domain and r^- is a negative domain for the *i*-th sample. The ranking loss is:

$$\mathcal{L}_{\mathrm{rank}} = -\frac{1}{|\mathcal{P}|}\sum_{(i,r^+,r^-)\in\mathcal{P}}\log\sigma\left(\tilde{z}_{ir^+}^{\mathrm{rec}} - \tilde{z}_{ir^-}^{\mathrm{rec}}\right) \tag{25}$$

This term encourages an applicable rehabilitation domain to receive a higher score than a non-applicable domain for the same sample.

The three supervised losses are combined through Eq. (6). The classification term addresses cognitive-status discrimination, the multi-label term estimates domain applicability under label imbalance, and the ranking term improves the relative order of the resulting domain scores.

4.7 Transfer and optimization strategy

Expert-endorsed rehabilitation-domain labels require specialist review and are more costly to obtain than

unannotated assessment profiles. The shared encoder is therefore initialized using the parameters learned during self-supervised pre-training.

Let $\theta_{\text{enc}}^{\text{pre}}$ denote the pre-trained encoder parameters. The initial parameters of the downstream encoder are set as:

$$\theta_{\text{enc}}^{\text{ft},0} \leftarrow \theta_{\text{enc}}^{\text{pre}} \tag{26}$$

where $\theta_{\text{enc}}^{\text{ft},0}$ denotes the encoder state at the beginning of supervised fine-tuning.

The encoder parameters are divided into frozen and trainable subsets:

$$\theta_{\text{enc}} = \{\theta_{\text{frozen}}, \theta_{\text{trainable}}\}, \quad \nabla_{\theta_{\text{frozen}}} \mathcal{L}_{\text{fine}} = 0 \tag{27}$$

Selected lower Transformer blocks remain fixed during fine-tuning, whereas the remaining encoder layers, the two task-specific branches, and the prior-fusion coefficient are optimized using the supervised objective. This strategy retains part of the representation learned during pre-training while allowing higher-level features to adapt to the two downstream tasks.

The empirical prior matrix is constructed once from the training labels before supervised optimization. It is not updated through backpropagation. In contrast, η is learned jointly with the trainable encoder parameters and task-specific heads. **Algorithm 1** summarizes the training procedure of CMT-Transformer.

Algorithm 1: Training Procedure of CMT-Transformer

Input: Training data $\mathcal{D}_{\text{train}}$, encoder f_θ, reconstruction decoder g_ψ, classification head g_{cls}, rehabilitation-domain prediction head g_{rec}, and hyperparameters λ_c, α, β, δ, τ.

Output: Cognitive-status probabilities $\mathbf{p}^{\text{cls}}$ and rehabilitation-domain probabilities $\hat{\mathbf{y}}^{\text{rec}}$.

Stage 1: Self-supervised encoder initialization

1: Initialize encoder parameters θ^{pre} and decoder parameters ψ.

2: **for** each pre-training minibatch $\mathbf{X}$ **do**

3: Generate a masked input $\tilde{\mathbf{X}}$ and two augmented views $\tilde{\mathbf{X}}^{(1)}$, $\tilde{\mathbf{X}}^{(2)}$.

4: Encode the two augmented views and reconstruct the masked input using $f_{\theta^{pre}}$ and g_ψ.

5: Compute $\mathcal{L}_{\text{recon}}$ and $\mathcal{L}_{\text{contrast}}$ using Eqs. (9) and (12).

6: Update θ^{pre} and ψ using $\mathcal{L}_{\text{pre}}$.

7: **end for**

Stage 2: Assessment-conditioned multi-task fine-tuning

8: Initialize the downstream encoder using Eq. (26), and freeze the selected lower Transformer blocks according to Eq. (27).

9: Construct the empirical assessment-rehabilitation prior matrix $\mathbf{P}$ from the training labels using Eq. (20).

10: **for** each supervised minibatch $\left\{\left(\mathbf{x}_i, y_i^{\text{cls}}, \mathbf{y}_i^{\text{rec}}\right)\right\}_{i=1}^{B}$ **do**

11: Construct the feature-token sequence using Eqs. (13) and (14), and obtain the global representation $\mathbf{h}$.

12: Compute the cognitive-status logits and probabilities using Eq. (17).

13: Construct the assessment-conditioned representation and compute the base rehabilitation-domain logits using Eqs. (18) and (19).

14: Fuse the empirical prior and compute the rehabilitation-domain probabilities using Eqs. (21) and (22).

15: Compute $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{ml}}$, $\mathcal{L}_{\text{rank}}$.

16: Update all trainable parameters using $\mathcal{L}_{\text{fine}}$.

17: **end for**

18: Return $\mathbf{p}^{\text{cls}}$, $\hat{\mathbf{y}}^{\text{rec}}$.

4.8 Computational complexity analysis

Let $n = d + 1$ denote the number of input tokens including the [CLS] token, $D = d_{\text{model}}$ the embedding dimension, D_{ff} the hidden dimension of the feed-forward network, L the number of Transformer layers, and B the minibatch size. The time complexity of the Transformer backbone is $\mathcal{O}\left(BL(n^2 D + nD^2 + nDD_{\text{ff}})\right)$, where the three terms correspond to self-attention, linear projections, and feed-forward computation, respectively. During self-supervised pre-training, masked reconstruction introduces an additional cost of approximately $\mathcal{O}(B|\mathcal{M}|D)$, while batch-wise contrastive similarity computation requires $\mathcal{O}(B^2 D)$. The assessment-rehabilitation prior fusion requires only $\mathcal{O}(BCR)$, where C and R are the numbers of assessment categories and rehabilitation domains, and therefore contributes limited overhead relative to the Transformer encoder. The principal activation-memory complexity during training is approximately $\mathcal{O}(BL(nD + n^2))$, excluding model parameters and optimizer states. Freezing the lower encoder layers reduces gradient storage, optimizer-state memory, and backward-pass computation, although it does not change the asymptotic forward-pass complexity.

5. Experiments

5.1 Dataset and experimental setup

This study used data from the China Health and Retirement Longitudinal Study (CHARLS), a publicly available longitudinal survey (https://charls.pku.edu.cn/) [37]. CHARLS was approved by the Biomedical Ethics Review Committee of Peking University (No. IRB00001052-11015), and informed consent was obtained during the original survey. The present study involved secondary analysis of publicly available de-identified data. A total of 5,099 participants aged 60 years or above were retained after excluding records with incomplete demographic information, missing cognitive-assessment variables, or unavailable rehabilitation-domain annotations. Cognitive status was divided into four categories based on the adopted Mini-Mental State Examination (MMSE) criteria: normal cognition, mild cognitive impairment, moderate cognitive impairment, and severe cognitive impairment.

The input profile contained demographic variables, including age, sex, nationality, and education level, together with item-level or subscale information from the MMSE, Telephone Interview for Cognitive Status, Animal Category Fluency task, and Community Screening Instrument for Dementia. The aggregate MMSE score used to define the cognitive-status categories was excluded from the model input. Accordingly, the classification task was treated as cognitive-status stratification from constituent assessment information rather than as an independent clinical diagnosis.

Rehabilitation-domain reference annotations were independently assigned by two senior physicians with more than five years of experience in cognitive rehabilitation from a tertiary-level hospital in China. The six annotated domains were orientation, attention, calculation, memory, executive function, and language training. Disagreements were resolved by consensus. These annotations represented potentially applicable rehabilitation domains based on the available structured profiles and did not indicate observed treatment

response or therapeutic benefit.

The dataset was divided into training, validation, and test sets at a ratio of 60:20:20. During self-supervised pre-training, the feature-masking ratio was set to 0.15. During supervised fine-tuning, selected lower encoder layers were frozen, and the loss weights for cognitive-status classification, multi-label domain prediction, and pairwise ranking were set to 0.6, 0.3, and 0.1, respectively.

All models were trained for 50 epochs using AdamW with a batch size of 64, a learning rate of 1×10^{-4}, and a weight decay of 2×10^{-2}. Transformer-based models used an embedding dimension of 128, a feed-forward dimension of 512, four attention heads, and a dropout rate of 0.12. Hyperparameters and model checkpoints were selected using the validation set. Experiments were implemented in PyTorch 2.7 and conducted on an NVIDIA RTX 5060 GPU. The reported results were averaged over ten independent runs. **Table 2** summarizes the structural configurations and distinctive modules of all compared methods.

Table 2 Structural configurations and distinctive modules of compared models.

Models	Parameter settings
VIME	Self- and semi-supervised representation learning with masked feature prediction; mask probability P_m = 0.5; Gaussian noise std = 0.01
SCARF	Contrastive pre-training via random feature corruption; corruption ratio = 0.6; Gaussian noise std = 0.008
TransTab	Transformer encoder with gated attention and column-wise tokenization; number of bins = 16
MMoE	Multi-gate Mixture-of-Experts with 4 experts; shared expert gating for task-specific learning; Gaussian noise std = 0.008
UniTTab	Unified Transformer for tabular data; frequency encoding L=8; feature fusion via temporal attention
TUTA	Tree-based Transformer with hierarchical structure embedding; distance threshold = 4
FT-Transformer	Feature-tokenized Transformer with PreNorm structure; layer normalization before each attention block
CMT-Transformer	Feature-tokenized Transformer with masked reconstruction and contrastive initialization; probabilistic assessment conditioning and learnably weighted empirical prior integration η initialized at 0.5

5.2 Baselines and evaluation metrics

To evaluate the effectiveness of CMT-Transformer, seven representative deep learning models for tabular data were compared. These include self-supervised learning models (VIME [15], SCARF [16]), transformer representation models (TransTab [17], TUTA [38], FT-Transformer [13]), multi-task structure models (MMoE [18]), and unified attention-based tabular models (UniTTab [39]). These methods cover complementary modeling strategies, including masked feature prediction, contrastive learning, cross-table transfer, structural attention, feature tokenization, and multi-task parameter sharing.

For methods that did not natively support the two-task setting, the original representation backbone was retained and connected to matched cognitive-status classification and rehabilitation-domain prediction heads. MMoE retained its expert-gating structure for task-specific learning. All methods used the same data partitions, output dimensions, evaluation protocol, and training budget. Model-specific pre-training and architectural components were retained where applicable.

Cognitive-status classification was evaluated using accuracy (ACC), weighted F1-score (F1(w)), and the area under the receiver operating characteristic curve under the one-vs-rest setting (AUC-OVR). ACC measures the overall proportion of correctly classified samples, while F1(w) accounts for class-specific precision and recall according to class frequency. AUC-OVR evaluates the discrimination of each cognitive-status category against the remaining categories.

Rehabilitation-domain prediction was evaluated using mean average precision (mAP), the Jaccard coefficient, and Hamming loss. mAP measures the ranking quality of the predicted domain scores, the Jaccard coefficient evaluates the overlap between the predicted and reference domain sets, and Hamming loss measures the proportion of incorrectly assigned labels.

Precision@3 (P@3), Recall@3 (R@3), and normalized discounted cumulative gain at three (nDCG@3) were additionally reported to evaluate the three highest-ranked rehabilitation-domain outputs. P@3 measures the proportion of relevant domains among the top three predictions, R@3 measures the proportion of reference domains retrieved within the top three positions, and nDCG@3 accounts for both relevance and ranking position. Higher values indicate better performance for all reported metrics except Hamming loss, for which a lower value is preferred.

5.3 Main results

Table 3 shows the performance of each model on the test set across all evaluation metrics. CMT-Transformer achieved the strongest mean performance among the compared models across all reported metrics. In the cognitive-status classification task, the average ACC, F1(w), and AUC-OVR of CMT-Transformer reached 0.8899, 0.8896, and 0.9740, respectively, representing the highest average levels among all models. In terms of the best values across independent runs, CMT-Transformer reached 0.9108 in ACC and 0.9103 in F1(w). FT-Transformer obtained slightly higher best values of 0.9127 and 0.9119, respectively; however, CMT-Transformer achieved higher mean ACC and F1(w), indicating more consistently strong performance across runs. MMoE obtained the highest best-run AUC-OVR of 0.9826, whereas CMT-Transformer achieved the highest mean AUC-OVR with a relatively low standard deviation. In the multi-label rehabilitation-domain prediction task, the average values for mAP, Jaccard, P@3, R@3, and nDCG@3 reached 0.9535, 0.7217, 0.7292, 0.7756, and 0.8526, respectively, indicating stronger mean rehabilitation-domain prediction and ranking performance than the compared baselines. Meanwhile, the mean and best values of Hamming loss decreased to 0.0866 and 0.0846, respectively, indicating a lower multi-label prediction error rate. Overall, CMT-Transformer showed competitive performance and relatively stable results across most metrics, suggesting the combined benefit of self-supervised pre-training, probabilistic conditioning and empirical prior integration.

Table 3 Overall performance comparison of baseline models across all evaluation metrics.

Metric	Statistic	VIME	SCARF	TransTab	MMoE	UniTTab	TUTA	FT-Transformer	CMT-Transformer
	Std	0.0117	0.0111	0.0115	**0.0066**	0.0120	0.0088	0.0222	0.0111
ACC	Mean	0.8818	0.8773	0.8857	0.8823	0.8523	0.8811	0.8728	**0.8899**
	Best	0.8980	0.9000	0.9029	0.8931	0.8755	0.8951	**0.9127**	0.9108
	Std	0.0117	0.0112	0.0115	**0.0065**	0.0124	0.0090	0.0216	0.0112
F1(w)	Mean	0.8815	0.8768	0.8855	0.8826	0.8516	0.8811	0.8727	**0.8896**
	Best	0.8981	0.8997	0.9029	0.8933	0.8747	0.8953	**0.9119**	0.9103
	Std	0.0045	0.0045	0.0046	0.0042	0.0059	0.0037	0.0083	**0.0034**
AUC-OVR	Mean	0.9696	0.9684	0.9704	0.9738	0.9512	0.9711	0.9663	**0.9740**
	Best	0.9786	0.9783	0.9760	**0.9826**	0.9633	0.9761	0.9825	0.9786
	Std	0.0054	0.0053	0.0066	0.0081	0.0109	0.0059	0.0254	**0.0033**
mAP	Mean	0.9400	0.9454	0.9448	0.9180	0.8960	0.9368	0.9053	**0.9535**
	Best	0.9469	0.9512	0.9523	0.9265	0.9155	0.9431	0.9274	**0.9584**
	Std	0.0136	0.0109	0.0110	0.0137	0.0296	0.0141	0.0306	**0.0092**
Jaccard	Mean	0.6944	0.7005	0.6998	0.6518	0.6288	0.6909	0.6534	**0.7217**
	Best	0.7131	0.7218	0.7201	0.6707	0.6730	0.7083	0.6959	**0.7289**
	Std	0.0065	0.0065	0.0064	0.0067	0.0092	0.0065	0.0154	**0.0045**
P@3	Mean	0.7223	0.7250	0.7251	0.7117	0.7030	0.7211	0.7085	**0.7292**
	Best	0.7304	0.7337	0.7343	0.7239	0.7206	0.7284	0.7275	**0.7350**
	Std	0.0100	0.0101	0.0095	0.0100	0.0111	0.0100	0.0181	**0.0082**
R@3	Mean	0.7691	0.7717	0.7716	0.7578	0.7490	0.7676	0.7555	**0.7756**
	Best	0.7812	0.7845	0.7825	0.7745	0.7643	0.7777	0.7745	**0.7859**
	Std	0.0082	0.0082	0.0086	0.0069	0.0124	0.0084	0.0171	**0.0064**
nDCG@3	Mean	0.8463	0.8494	0.8467	0.8270	0.8190	0.8421	0.8260	**0.8526**
	Best	0.8558	0.8598	0.8575	0.8348	0.8420	0.8522	0.8512	**0.8604**
	Std	0.0034	0.0033	0.0035	0.0051	0.0085	0.0055	0.0162	**0.0018**
Hamming	Mean	0.1029	0.0968	0.0962	0.1192	0.1408	0.1054	0.1295	**0.0866**
	Best	0.0987	0.0925	0.0912	0.1145	0.1216	0.0998	0.1098	**0.0846**

*The optimal results are shown in bold.

To visually compare the relative performance of each model, **Fig. 4** shows a row-normalized heatmap of the mean values in **Table 3**. Color intensity reflects the relative performance of each model on a specific metric; darker colors indicate better relative performance. For Hamming loss, lower values indicate better performance, and the values were normalized accordingly before visualization. The heatmap provides an intuitive comparison of model performance across classification and rehabilitation-domain prediction metrics, with CMT-Transformer showing consistently strong performance across most evaluation criteria.

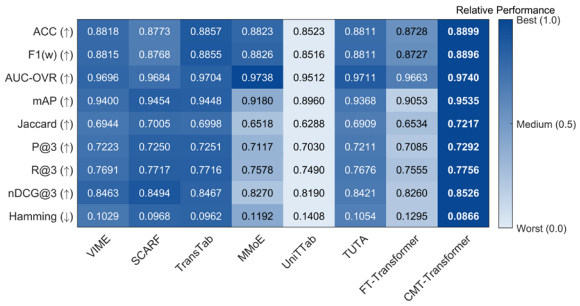


Fig. 4 Row normalized heatmap of experimental results (mean value).

Fig. 5 illustrates the convergence dynamics of various evaluation metrics on the validation set. In the cognitive impairment classification task, CMT-Transformer shows faster convergence and relatively stable validation performance. Observing the convergence curves of F1(w) and AUC-OVR, it can be seen that the model's performance rapidly climbs to a high level within the first 10 training epochs and maintains good stability in subsequent training epochs. In contrast, the performance improvement of other baseline models in the same stage is relatively slow, while UniTTab's F1(w) curve shows significant fluctuations. This indicates that CMT-Transformer can fit the classification boundary faster and more accurately, effectively overcoming the blind spots in the early training stage.

In terms of recommendation task-related metrics, the model also maintains a comprehensive and significant advantage. Observing the trends of mAP, Jaccard, and nDCG@3, the CMT-Transformer's curves not only converge smoothly but also consistently outperform all comparative methods throughout the entire 50 epochs of training. Its Hamming loss curve rapidly decreases to around 0.10 after approximately 20 epochs of training, and further approaches a minimum value of 0.08 in later stages. In contrast, the error of most baseline models decreases slowly, often stagnating above 0.10 in later stages. This further highlights its accuracy in multi-label ranking and combined prediction.

Overall, the CMT-Transformer exhibits excellent convergence characteristics across various evaluation metrics. It shows minimal fluctuations and no obvious signs of overfitting, closely matching the final quantitative results shown in the table. This fast and stable training dynamic fully demonstrates the advantages of combining self-supervised pre-training and conditional multi-task structure. It not only effectively improves the model's feature representation capabilities in small-sample scenarios but also significantly enhances its generalization performance, enabling it to handle classification and recommendation tasks more easily.

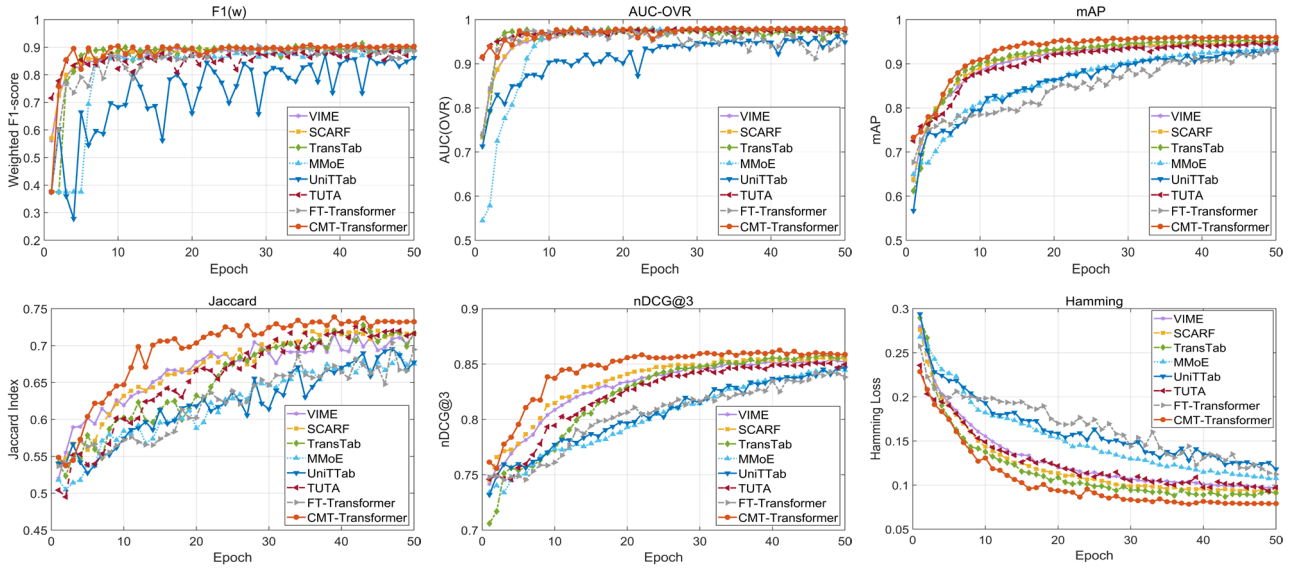


Fig. 5 Convergence curves of all models across evaluation metrics

Fig. 6 visually illustrates the convergence of each model in terms of training loss and validation loss. It can be seen that CMT-Transformer consistently outperforms all comparison models in both optimization speed and convergence quality. Specifically, its training loss exhibits a rapid decreasing trend in the first 10 epochs, steadily declining in the later stages and eventually stabilizing at the lowest level among all models (around 1.0). Simultaneously, its validation loss also decreases significantly in the early stages of training and eventually converges smoothly to the optimal position of approximately 0.8. In contrast, the loss decreases relatively slowly in other methods, accompanied by varying degrees of oscillation during convergence. In particular, FT-Transformer and UniTTab even exhibited drastic abnormal fluctuations on the validation set, showing significantly weaker overall performance than CMT-Transformer. Overall, the training and validation curves of CMT-Transformer remain highly smooth throughout, with no signs of overfitting such as loss bounce. This fully demonstrates that the combination of self-supervised pre-training and conditional multi-task structure can effectively improve the optimization efficiency and generalization ability of the model in small sample scenarios.

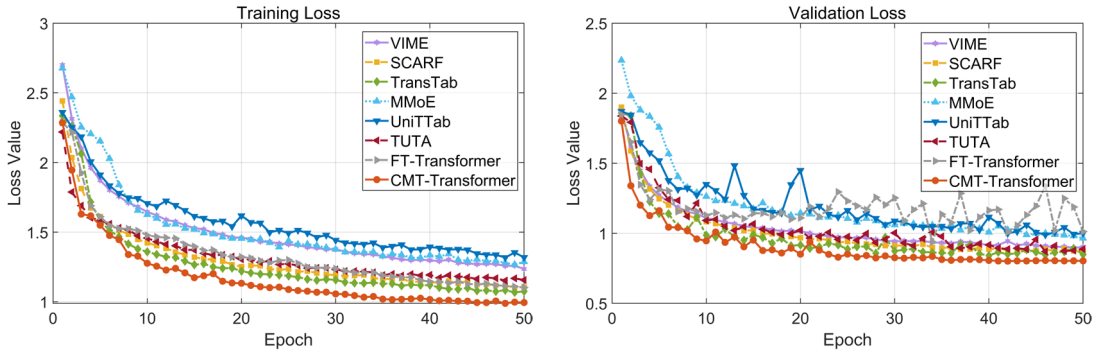


Fig. 6 Training and validation loss curves of all models.

As shown in **Fig. 7** and the corresponding time consumption statistics, under the same hardware conditions, the average running time of each model shows significant differences. TransTab has the highest training time

cost, more than twice that of lightweight baseline models such as MMoE and SCARF. TUTA and UniTTab also have relatively high time costs, clearly at a high level. In contrast, CMT-Transformer demonstrates excellent computational efficiency. Its average time consumption is not only significantly lower than the three high-time-consuming methods mentioned above, but also slightly better than FT-Transformer and VIME, which belong to the Transformer architecture. Although its training time is slightly increased compared to the simplest structures, MMoE and SCARF, it still firmly remains in the computational tier of lightweight models.

CMT-Transformer does not incur a heavy computational burden due to the introduction of self-supervised pre-training and conditional multi-task structures. While significantly improving model prediction accuracy and convergence stability, it successfully controls the time consumption to a level comparable to basic lightweight models, suggesting a favorable balance between performance and computational cost.

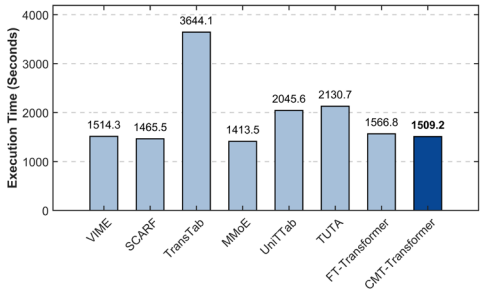


Fig. 7 Comparison of running time for different models.

5.4 Ablation study

The ablation experiment involves removing model components one by one to analyze the independent contribution and effectiveness of each module to the overall performance. **Table 4** shows the results of the ablation experiments presented in this paper. The results show that each key module contributes to the cognitive-status stratification and rehabilitation-domain prediction performance.

Among all variants, the full model achieved the best performance across the selected evaluation metrics. Removing the contrastive self-supervised branch resulted in a clear decline in performance, with F1(w) and AUC-OVR decreasing to 0.8718 and 0.9634, respectively. This result suggests that contrastive pre-training provides useful feature representations and contributes to cognitive-status stratification.

Removing the probabilistic conditioning module caused relatively small changes in classification-related metrics, while the rehabilitation-domain prediction metrics showed a more evident decline. Specifically, mAP and nDCG@3 decreased to 0.9326 and 0.8294, respectively, representing the lowest values among all variants. This finding indicates that transferring the cognitive-status probability distribution to the downstream branch plays an important role in modeling the relationship between assessment states and rehabilitation-domain prediction.

Removing the empirical prior integration component resulted in a relatively small decrease in overall classification performance, but reduced the Jaccard coefficient to 0.6948 and also decreased mAP and

nDCG@3 to 0.9416 and 0.8405, respectively. These results suggest that the training-derived assessment-rehabilitation prior provides complementary population-level information for improving multi-label consistency and domain ranking performance.

Table 4 Ablation experiment results of CMT-Transformer.

Variant	Contrastive objective	Probabilistic conditioning	Empirical prior integration	F1 (w)	AUC-OVR	mAP	Jaccard	nDCG@3
Full CMT-Transformer	√	√	√	0.8896	0.9740	0.9535	0.7217	0.8526
w/o Contrastive	-	√	√	0.8718	0.9634	0.9387	0.7058	0.8361
w/o Probabilistic Conditioning	√	-	√	0.8879	0.9731	0.9326	0.7019	0.8294
w/o Empirical Prior Integration	√	√	-	0.8890	0.9736	0.9416	0.6948	0.8405

5.5 Statistical test and stability

For statistical significance analysis, this study used a two-tailed paired-samples t-test [40, 41], because all models were compared under the same data partitioning, training scheme, and 10-run settings. This test was used to compare the performance of CMT-Transformer with each baseline model on nine assessment metrics. As shown in **Table 5**, most paired comparisons were significant at the 0.05 level, particularly for rehabilitation-domain prediction and ranking metrics such as mAP, P@3, and nDCG@3. However, ACC and F1(w) did not differ significantly from TransTab, F1(w) did not differ significantly from MMoE, and AUC-OVR did not differ significantly from MMoE. Overall, the results support performance differences for most comparisons, especially in rehabilitation-domain prediction and ranking metrics.

As shown in **Table 6**, the Friedman test [42] was used for the overall ranking analysis of the mean values. CMT-Transformer achieved the best average rank of 1.00 among all candidate models. TransTab and SCARF ranked second and third, with average ranks of 2.67 and 3.56, respectively. These ranking results are consistent with the metric-level comparisons reported above. By jointly considering the overall performance across multiple evaluation metrics, the ranking analysis further supports the effectiveness of the proposed CMT-Transformer framework.

Table 5 P-values of paired t-tests between CMT-Transformer and baseline models across evaluation metrics.

Metric	VIME	SCARF	TransTab	MMoE	UniTTab	TUTA	FT-Transformer
ACC	7.72E-03	2.32E-05	2.07E-01	4.55E-02	4.36E-07	1.50E-03	1.34E-02
F1(w)	8.93E-03	2.42E-05	2.24E-01	6.74E-02	6.58E-07	2.01E-03	1.12E-02
AUC-OVR	9.48E-04	3.52E-04	3.81E-04	7.41E-01	1.08E-08	1.59E-04	1.46E-02
mAP	8.97E-08	9.50E-06	1.85E-04	6.50E-09	1.63E-08	7.20E-08	4.48E-05
Jaccard	3.65E-05	4.80E-05	1.66E-04	2.37E-09	3.58E-06	2.57E-05	1.12E-05
P@3	1.71E-04	3.61E-03	1.53E-03	5.95E-07	1.48E-06	1.09E-04	1.66E-04
R@3	1.30E-04	3.20E-03	9.00E-04	5.02E-07	2.07E-06	9.38E-05	2.37E-04
nDCG@3	1.35E-04	1.38E-02	1.83E-04	1.61E-08	2.43E-06	1.31E-05	3.24E-05
Hamming	1.85E-08	2.67E-07	2.96E-06	1.22E-09	3.06E-09	1.49E-07	5.95E-06

Table 6 Friedman test results across evaluation metrics.

Algorithm	Mean Rank	Rank
CMT-Transformer	1.00	1
TransTab	2.67	2
SCARF	3.56	3
VIME	4.11	4
TUTA	4.78	5
MMoE	5.00	6
FT-Transformer	6.89	7
UniTTab	8.00	8

Table 7 presents a sensitivity analysis of three key hyperparameters in CMT-Transformer on the validation set. The selected setting, with a masking ratio of 0.15, a contrastive weight λ_c of 0.3, and a prior-fusion coefficient η of 0.5, achieved the best validation performance among the evaluated settings. Increasing the masking ratio or contrastive loss weight resulted in a slight performance degradation, suggesting that excessive perturbation or contrastive regularization may affect the preservation of informative assessment features. For prior fusion, η=0.5 produced better results than the other tested initializations, suggesting that an appropriate prior contribution can provide complementary guidance while maintaining the flexibility of data-driven prediction. Since η is jointly optimized during supervised fine-tuning, these results mainly reflect sensitivity to initialization rather than the effect of a fixed prior-fusion strength.

Table 7 Sensitivity analysis of key settings in CMT-Transformer.

Analysis factor	Setting	F1(w)	mAP	Jaccard	nDCG@3
Masking ratio	0.1	0.8835	0.9472	0.7071	0.8412
	0.15	0.8878	0.9506	0.7147	0.8465
	0.3	0.8792	0.9436	0.7015	0.8357
Contrastive weight (λ_c)	0	0.8724	0.9368	0.6982	0.8311
	0.3	0.8878	0.9506	0.7147	0.8465
	1.0	0.8809	0.9441	0.7037	0.8372
Initial prior-fusion coefficient (η)	0	0.8851	0.9428	0.6946	0.8345
	0.5	0.8878	0.9506	0.7147	0.8465
	1.0	0.8859	0.9457	0.7072	0.8401

To complement the statistical significance analysis, **Fig. 8** shows the distributions of six core test-set metrics across ten independent runs. Each point denotes the final test-set result of one run. The boxplots report the median, interquartile range, and run-to-run variation for each model [43, 44], with CMT-Transformer highlighted for comparison. The distributions are generally consistent with the mean, standard deviation, and best values reported in Table 3. CMT-Transformer shows stable performance on both cognitive-status classification and rehabilitation-domain prediction metrics, especially for mAP,

Jaccard, and nDCG@3. These results suggest that its performance advantage is not caused by a single favorable run, but is maintained across repeated experiments.

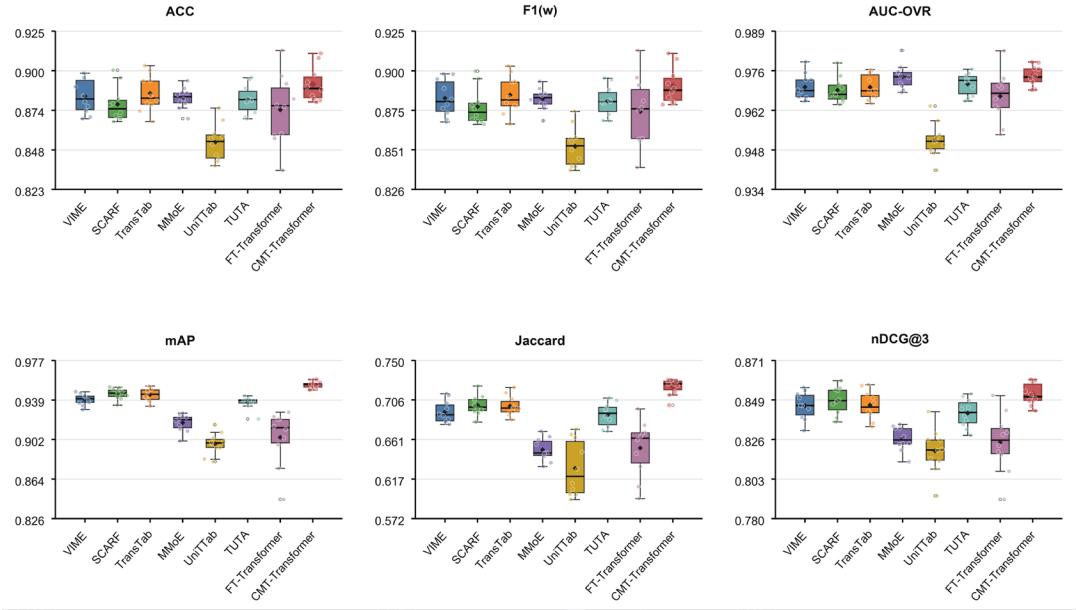


Fig. 8 Boxplot comparison of six core test-set metrics across ten independent runs.

5.6 Confusion matrix and error-pattern analysis

Fig. 9 presents the confusion matrix from a representative held-out test run, which achieved an ACC of 0.8814. The results show a clear diagonal pattern across the four cognitive-status levels, suggesting good agreement between the predicted and reference labels.

Most errors occur between adjacent severity levels. The main confusion is observed between normal cognition and mild cognitive impairment. This is expected, as the boundary between healthy aging and early mild cognitive impairment is often gradual and may be difficult to distinguish using structured assessment profiles alone. Importantly, severe cross-level errors are rare. Specifically, the number of cases in which a true severe cognitive impairment case was classified as normal is 0, and the number of normal cases classified as severe is also 0. These results suggest that CMT-Transformer tends to produce adjacent-level errors rather than severe category shifts. This analysis should be interpreted as a test-set error-pattern assessment rather than direct evidence of clinical diagnostic safety.

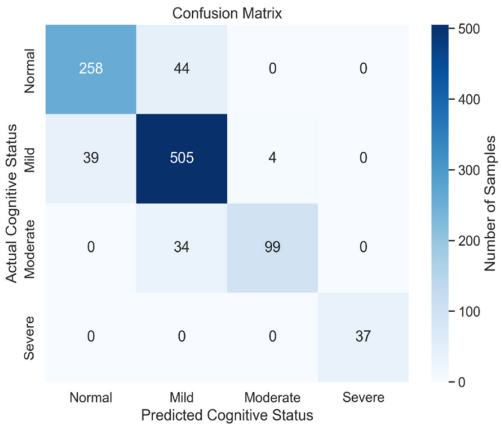


Fig. 9 Confusion matrix of CMT-Transformer on the held-out test set.

5.7 Domain-level prediction performance

In addition to the aggregate multi-label results, **Fig. 10** reports average precision (AP) for the six annotated rehabilitation domains: orientation, attention, calculation, memory, executive function, and language training.

The model achieved comparatively high AP values across the six domains, although performance varied by domain. Calculation and memory obtained relatively high AP values. This may reflect stronger alignment between their expert-endorsed reference annotations and the item-level variables available in the structured assessment profiles. Orientation, language training, and executive function also showed comparatively high AP values.

Attention obtained a lower AP value than the other domains. One possible explanation is that the available structured questionnaire and scale variables provided less consistent separation for this domain. These interpretations are descriptive and would require confirmation using independent cohorts or richer behavioral and physiological information.

Overall, the domain-level analysis indicates that CMT-Transformer can produce differentiated prediction scores across the six annotated domains, while also showing that some domains may benefit from additional input information.

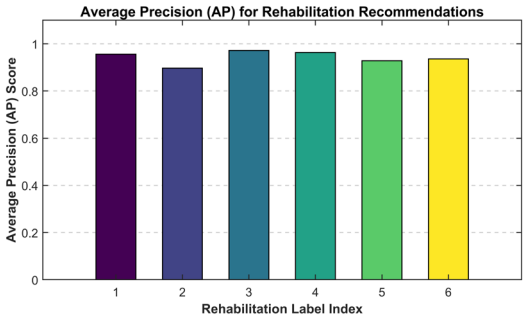


Fig. 10 Average precision of CMT-Transformer across six annotated rehabilitation domains on the held-out test set. Labels: 1–orientation, 2–attention, 3–calculation, 4–memory, 5–executive function, 6–language training.

5.8 System demonstration and illustrative case

To examine the potential applicability of the proposed framework, an interactive web-based decision-support system was developed, as shown in **Fig. 11**. The system was implemented in the Python ecosystem using a lightweight Streamlit browser/server architecture. It combines data input, model inference, and result visualization in a single workflow. The interface serves as a technical demonstration of how the trained framework can be incorporated into an interactive cognitive-assessment workflow.

During data entry and review, users can enter participant demographic information and edit cognitive-scale sub-items. For the representative case, the system constructs a structured assessment profile and performs cognitive-status stratification and rehabilitation-domain prediction.

Using the trained CMT-Transformer, the system displays the outputs of the two connected tasks. The predicted probability of mild cognitive impairment was 0.53. The complete cognitive-status probability

vector was then used as the conditional signal for the rehabilitation-domain branch. Orientation, executive function, and language training were the three highest-scoring domain outputs in this illustrative case.

The system displays the complete inference pathway from structured profile entry to cognitive-status probabilities and domain-level prediction scores. It does not automatically prescribe an intervention or replace professional assessment. The case should be regarded as a system-level illustration rather than a clinically validated deployment. Prospective evaluation in real clinical workflows remains necessary.

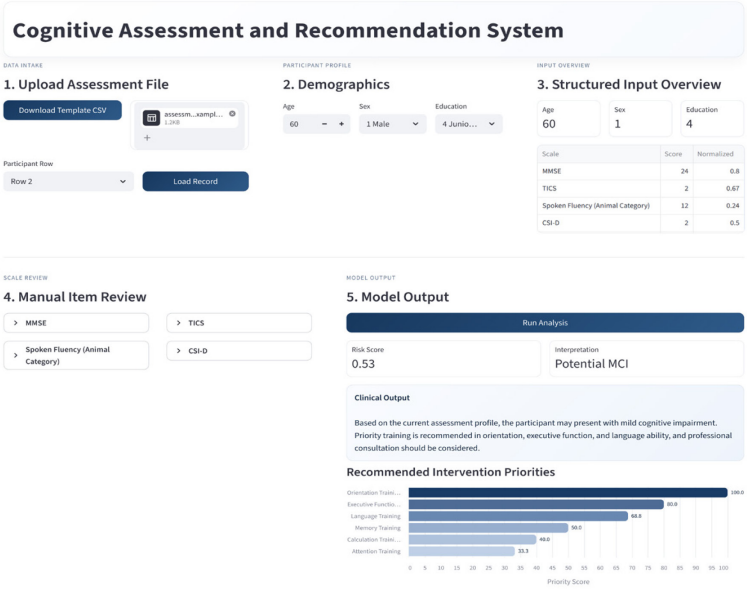


Fig. 11 Web-based decision-support system for cognitive-status stratification and rehabilitation-domain prediction based on CMT-Transformer.

6 Conclusions

This study proposed CMT-Transformer, a probabilistic task-coupled framework for joint cognitive-status stratification and rehabilitation-domain prediction from structured assessment profiles. The framework combines masked feature reconstruction and contrastive pre-training with a feature-tokenized Transformer encoder. Instead of treating the two tasks as parallel outputs, the complete cognitive-status probability distribution is supplied to the downstream branch as a differentiable conditioning signal. A training-derived assessment-rehabilitation prior is further projected through this distribution and incorporated into the rehabilitation-domain logits using a learnable fusion coefficient.

Experiments on a CHARLS-derived cohort of 5,099 older adults showed that CMT-Transformer achieved the strongest mean performance among the compared methods across the main classification, multi-label, and ranking metrics. The model obtained a mean ACC of 0.8899 for cognitive-status stratification and a mean mAP of 0.9535 for rehabilitation-domain prediction. Ablation experiments indicated that contrastive initialization mainly contributed to cognitive-status classification, whereas probabilistic conditioning and empirical prior integration had clearer effects on multi-label agreement and domain ranking. Statistical comparisons, repeated-run analyses, and sensitivity experiments further characterized the performance differences and model stability. A web-based system was also developed to

illustrate the complete inference workflow from structured assessment input to cognitive-status probabilities and domain-level prediction scores.

Several limitations should be considered. The model was evaluated on a single cohort, and the rehabilitation-domain labels were expert-endorsed references derived from structured profiles rather than observed intervention outcomes. The empirical prior reflected associations in the training data and should not be interpreted as causal clinical knowledge. In addition, the system has not been evaluated prospectively in clinical practice. Overall, CMT-Transformer provides a practical computational approach for modeling the directional relationship between cognitive-status assessment and rehabilitation-domain prediction. Future work will explore multimodal clinical data integration [45, 46], more sophisticated rehabilitation pathway modeling [47, 48], and cross-institutional transfer learning [49, 50].

References

1. Hu M, Shu X, Yu G, et al (2021) A Risk Prediction Model Based on Machine Learning for Cognitive Impairment Among Chinese Community-Dwelling Elderly People With Normal Cognition: Development and Validation Study. J Med Internet Res 23:e20298. https://doi.org/10.2196/20298
2. Huo C-C, Zheng Y, Lu W-W, et al (2021) Prospects for intelligent rehabilitation techniques to treat motor dysfunction. Neural Regener Res 16:264. https://doi.org/10.4103/1673-5374.290884
3. Fong J, Ocampo R, Gross DP, Tavakoli M (2020) Intelligent Robotics Incorporating Machine Learning Algorithms for Improving Functional Capacity Evaluation and Occupational Rehabilitation. J Occup Rehabil 30:362–370. https://doi.org/10.1007/s10926-020-09888-w
4. Liu Q, Vaci N, Koychev I, et al (2022) Personalised treatment for cognitive impairment in dementia: development and validation of an artificial intelligence model. BMC Med 20:45. https://doi.org/10.1186/s12916-022-02250-2
5. Dipietro L, Eden U, Teixeira P, et al (2025) A multimodal machine learning approach to predict Fugl-Meyer scores and motor recovery potential in stroke rehabilitation: Toward precision-based therapies. Inf Sci 721:122564. https://doi.org/10.1016/j.ins.2025.122564
6. Zhang X, Jin F, Hu J, Xu J (2024) AI-Driven Health Monitoring: Integrating Transformer and Convolutional Fusion for Stroke Patient Posture Estimation. IEEE J Biomed Health Inform 1–11. https://doi.org/10.1109/JBHI.2024.3501353
7. Krishnan R, Rajpurkar P, Topol EJ (2022) Self-supervised learning in medicine and healthcare. Nat Biomed Eng 6:1346–1352. https://doi.org/10.1038/s41551-022-00914-1
8. Chen T, Kornblith S, Norouzi M, Hinton G (2020) A Simple Framework for Contrastive Learning of Visual Representations. In: Proceedings of the 37th International Conference on Machine Learning. PMLR, pp 1597–1607
9. Huang S-C, Pareek A, Jensen M, et al (2023) Self-supervised learning for medical image

classification: a systematic review and implementation guidelines. npj Digital Med 6:74. https://doi.org/10.1038/s41746-023-00811-0

10. He K, Fan H, Wu Y, et al (2020) Momentum Contrast for Unsupervised Visual Representation Learning. pp 9729–9738
11. Yang J, Gupta A, Upadhyay S, et al (2022) TableFormer: Robust Transformer Modeling for Table-Text Encoding. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Dublin, Ireland, pp 528–537
12. Somepalli G, Schwarzschild A, Goldblum M, et al (2022) SAINT: improved neural networks for tabular data via row attention and contrastive pre-training
13. Gorishniy Y, Rubachev I, Khrulkov V, Babenko A (2021) Revisiting Deep Learning Models for Tabular Data. In: Advances in Neural Information Processing Systems. Curran Associates, Inc., pp 18932–18943
14. Tu P, Li J, Wang H (2025) A Multitask Deep Learning Approach for sEMG-Based Human Motion Intention and Muscle Fatigue Levels Recognition. IEEE Trans Instrum Meas 74:1–10. https://doi.org/10.1109/TIM.2025.3550611
15. Yoon J, Zhang Y, Jordon J, van der Schaar M (2020) VIME: Extending the Success of Self- and Semi-supervised Learning to Tabular Domain. In: Advances in Neural Information Processing Systems. Curran Associates, Inc., pp 11033–11043
16. Bahri D, Jiang H, Tay Y, Metzler D (2021) Scarf: Self-Supervised Contrastive Learning using Random Feature Corruption
17. Wang Z, Sun J (2022) TransTab: Learning Transferable Tabular Transformers Across Tables. Adv Neural Inf Process Syst 35:2902–2915
18. Ma J, Zhao Z, Yi X, et al (2018) Modeling task relationships in multi-task learning with multi-gate mixture-of-experts. In: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. Association for Computing Machinery, New York, NY, USA, pp 1930–1939
19. Li J, Xue J, Cao R, et al (2024) FineRehab: A Multi-modality and Multi-task Dataset for Rehabilitation Analysis. In: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). IEEE, Seattle, WA, USA, pp 3184–3193
20. Kazuma H, caiming xiong, Yoshimasa T, Richard S (2017) A Joint Many-Task Model: Growing a Neural Network for Multiple NLP Tasks. Proc 2017 Conf Empir Methods Nat Lang Process 1923–1933. https://doi.org/10.18653/v1/d17-1206
21. Read J, Pfahringer B, Holmes G, Frank E (2011) Classifier chains for multi-label classification. Mach Learn 85:333–359. https://doi.org/10.1007/s10994-011-5256-5

22. Xia Y, Tong Y, Yang J, et al (2025) Multi-Label Prototype-Aware Structured Contrastive Distillation. Tsinghua Sci Technol 30:1808–1830. https://doi.org/10.26599/TST.2024.9010182
23. Dar SUH, Yurt M, Shahdloo M, et al (2020) Prior-Guided Image Reconstruction for Accelerated Multi-Contrast MRI via Generative Adversarial Networks. IEEE J Sel Top Signal Process 14:1072–1087. https://doi.org/10.1109/JSTSP.2020.3001737
24. Dai Z, Yi J, Yan L, et al (2023) PFEMed: Few-shot medical image classification using prior guided feature enhancement. Pattern Recognition 134:109108. https://doi.org/10.1016/j.patcog.2022.109108
25. Tian Z, Zhao H, Shu M, et al (2022) Prior Guided Feature Enrichment Network for Few-Shot Segmentation. IEEE Trans Pattern Anal Mach Intell 44:1050–1065. https://doi.org/10.1109/TPAMI.2020.3013717
26. Lienen J, Hüllermeier E (2021) From Label Smoothing to Label Relaxation. Proc AAAI Conf Artif Intell 35:8583–8591. https://doi.org/10.1609/aaai.v35i10.17041
27. Ridnik T, Ben-Baruch E, Zamir N, et al (2021) Asymmetric Loss For Multi-Label Classification. In: 2021 IEEE/CVF International Conference on Computer Vision (ICCV). IEEE, Montreal, QC, Canada, pp 82–91
28. Ding J, Yu G, He X, et al (2021) Sampler Design for Bayesian Personalized Ranking by Leveraging View Data. IEEE Trans Knowl Data Eng 33:667–681. https://doi.org/10.1109/TKDE.2019.2931327
29. Nicora G, Pe S, Santangelo G, et al (2025) Systematic review of AI/ML applications in multi-domain robotic rehabilitation: trends, gaps, and future directions. J NeuroEngineering Rehabil 22:79. https://doi.org/10.1186/s12984-025-01605-z
30. Zhang Y, Wu X, Fang Q, et al (2023) Knowledge-Enhanced Attributed Multi-Task Learning for Medicine Recommendation. ACM Trans Inf Syst 41:1–24. https://doi.org/10.1145/3527662
31. Gui J, Chen T, Zhang J, et al (2024) A Survey on Self-Supervised Learning: Algorithms, Applications, and Future Trends. IEEE Trans Pattern Anal Mach Intell 46:9052–9071. https://doi.org/10.1109/TPAMI.2024.3415112
32. Khosla P, Teterwak P, Wang C, et al (2020) Supervised Contrastive Learning. In: Advances in Neural Information Processing Systems. Curran Associates, Inc., pp 18661–18673
33. Liang G, Greenwell C, Zhang Y, et al (2022) Contrastive Cross-Modal Pre-Training: A General Strategy for Small Sample Medical Imaging. IEEE J Biomed Health Inform 26:1640–1649. https://doi.org/10.1109/JBHI.2021.3110805
34. Lin P-J, Zhai X, Li W, et al (2022) A Transferable Deep Learning Prognosis Model for Predicting Stroke Patients' Recovery in Different Rehabilitation Trainings. IEEE J Biomed Health Inform 26:6003–6011. https://doi.org/10.1109/JBHI.2022.3205436
35. Zhuang F, Qi Z, Duan K, et al (2021) A Comprehensive Survey on Transfer Learning. Proc IEEE 109:43–76. https://doi.org/10.1109/JPROC.2020.3004555

36. Gao F, Yoon H, Wu T, Chu X (2020) A feature transfer enabled multi-task deep learning model on medical imaging. Expert Syst Appl 143:112957. https://doi.org/10.1016/j.eswa.2019.112957
37. Zhao Y, Hu Y, Smith JP, et al (2014) Cohort Profile: The China Health and Retirement Longitudinal Study (CHARLS). Int J Epidemiol 43:61–68. https://doi.org/10.1093/ije/dys203
38. Wang Z, Dong H, Jia R, et al (2021) TUTA: Tree-based Transformers for Generally Structured Table Pre-training. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. ACM, Virtual Event Singapore, pp 1780–1790
39. Luetto S, Garuti F, Sangineto E, et al (2025) One transformer for all time series: representing and training with time-dependent heterogeneous tabular data. Mach Learn 114:145. https://doi.org/10.1007/s10994-025-06778-1
40. Rietveld T, van Hout R (2017) The paired *t* test and beyond: Recommendations for testing the central tendencies of two paired samples in research on speech, language and hearing pathology. J Commun Disord 69:44–57. https://doi.org/10.1016/j.jcomdis.2017.07.002
41. Alzahrani SM (2025) MS-trust: a transformer model with causal-global dual attention for enhanced MRI-based multiple sclerosis and myelitis detection. Complex Intell Syst 11:331. https://doi.org/10.1007/s40747-025-01945-2
42. Zhang Y, Du G, Tang H, et al (2026) Manifold-aware triple cooperative multi-population differential evolution with reinforcement learning for irregular 3D UAV path planning. Knowledge-Based Syst 341:115863. https://doi.org/10.1016/j.knosys.2026.115863
43. Zhou T, Wang Z, Liu X, et al (2025) Learning deep feature representations for multi-modal MR brain tumor segmentation. Neurocomputing 638:130162. https://doi.org/10.1016/j.neucom.2025.130162
44. Huang J, Li D, Yang P, et al (2026) Riemannian spatio-temporal graph neural network for enhanced cognitive load detection using EEG. Neurocomputing 685:133650. https://doi.org/10.1016/j.neucom.2026.133650
45. Steyaert S, Pizurica M, Nagaraj D, et al (2023) Multimodal data fusion for cancer biomarker discovery with deep learning. Nat Mach Intell 5:351–362. https://doi.org/10.1038/s42256-023-00633-5
46. Wang C, Tian Y, Yin S, et al (2026) Integrating multimodal clinical data with a large model for prostate cancer diagnosis. npj Digital Med 9:497. https://doi.org/10.1038/s41746-026-02670-x
47. Li M, He J, Yu H, et al (2024) A Clinical Pathway Model for Severe Stroke Rehabilitation based on Process Mining. In: 2024 World Rehabilitation Robot Convention (WRRC). pp 1–6
48. Jiao S, Huo H, Liu W, et al (2025) Gaitformer: a spatial-temporal attention-enhanced network without softmax for parkinson's disease early detection. Complex Intell Syst 11:236. https://doi.org/10.1007/s40747-025-01830-y
49. Rusak F, Koopman B, Brown NJ, et al (2023) Catching Misdiagnosed Limb Fractures in the

Emergency Department Using Cross-institution Transfer Learning. In: Muresan S, Chen V, Casey K, et al (eds) Proceedings of the 21st Annual Workshop of the Australasian Language Technology Association. Association for Computational Linguistics, Melbourne, Australia, pp 78–87

50. Li S, Li X, Yu K, et al (2025) Bridging Data Gaps in Healthcare: A Scoping Review of Transfer Learning in Structured Data Analysis. Health Data Sci 5:0321. https://doi.org/10.34133/hds.0321